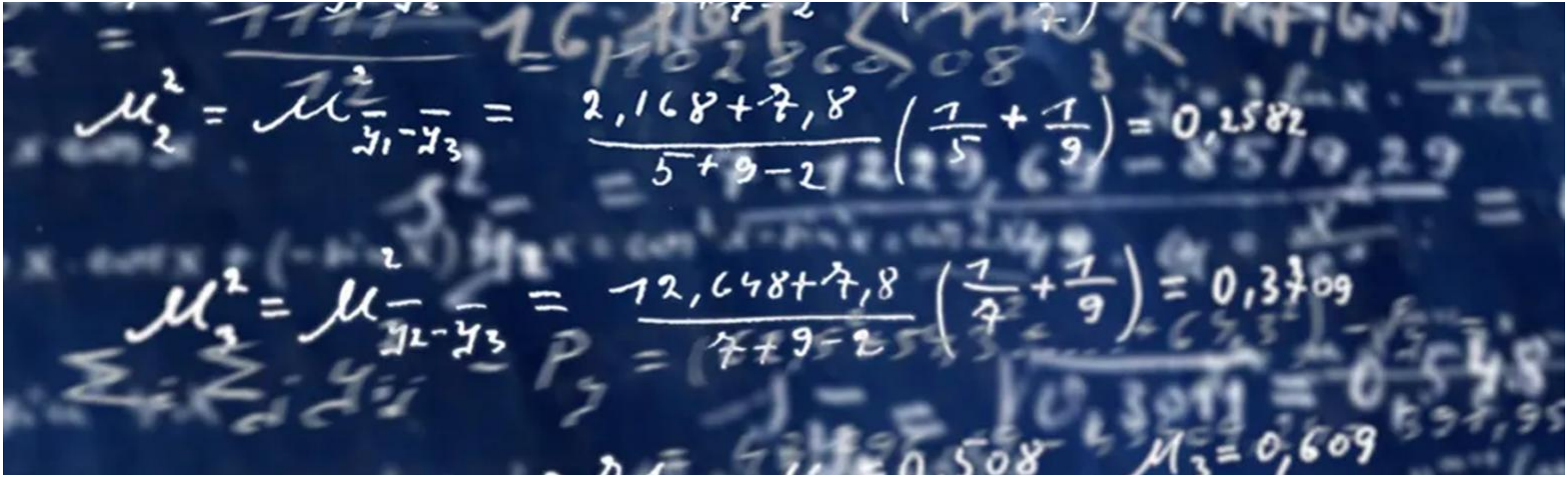

Predictive Analytics · Lecture 12

SUMMING IT ALL UP

CBS
COPENHAGEN BUSINESS SCHOOL
HANDELSHØJSKOLEN

F26

AGENDA

- Exam project
- Methodological round-up
- A final example – US house prices

THE EXAM PROJECT

IN ESSENCE

- Deadline for hand-in of report: **29 May 2026 (12:00)**, group hand-in on *Digital Exam*
- Length of report: **15** page essay (we will not read anything that exceeds 15 pages)
- Group: **3–4** group members
- Oral exam: **16 – 19 June 2026**
- Requirements:
  - Find and use your own time series data.
  - Analyze your time series with at least 2 methods among the ones presented in the course.
  - Produce forecasts and discuss results.
    (*Based on statistical methods, i.e., not judgmental forecasts*)
  - The report should be self-explanatory: Everything you consider relevant should be in the main body (output, figures, tables). The code can be placed in the appendix, but this is not essential.
  - Use R for analysis.

Window before the exam: You can email teachers in the course if you have questions about the exam. However, we will **not reply to emails the last two weeks** before the deadline.

SV: Predictive Project Deadline and NLP Oral Exams Overlap - Message (HTML)

SV: Predictive Project Deadline and NLP Oral Exams Overlap

Maja Jeziorski Nielsen
To Herdis Steingrimsdottir; Thomas Einfeldt

ti 05-05-2026 14:01

Hi Herdis and Thomas,

Thank you for your email.

I understand the students concern, but this is not an overlap.

They may submit their exam paper latest on the 29th of May, thus they can certainly submit it earlier.

We unfortunately cannot change the exam dates once they have been published.

Best,
Maja

Fra: Herdis Steingrimsdottir <hs.eco@cbs.dk>
Sendt: 5. maj 2026 13:30
Til: Thomas Einfeldt <tce.eco@cbs.dk>; Maja Jeziorski Nielsen <mjn.stu@cbs.dk>
Emne: FW: Predictive Project Deadline and NLP Oral Exams Overlap

Hi Maja, cc Thomas,
Should we try to accommodate the students (see below) - or is that not possible at this point?
Kind regards,
Herdis

LEARNING OBJECTIVES

To achieve the grade of 12, students should meet the following learning objectives only with no or minor mistakes or errors. By the end of the course the students will be able to:

1. **Explain** the concepts, models and methods introduced during the course.
2. Identify and perform an **academically founded forecasting analysis in practice** (including taking relevant theoretical relationships into account) using the tools introduced during the course.
3. **Discuss and solve any important problems** encountered in relation to the analysis.
4. **Evaluate** the forecasts of the analysis.
5. **Report/communicate** the results and conclusion of the analysis both if the reader is a statistician/econometrician and if the reader is e.g. a CEO.
6. Evaluate a forecasting analysis **conducted by another** person/researcher.
7. **Use a statistical analysis software package** for the analysis (e.g. R or SAS) and demonstrate that you can interpret and evaluate the output from the software package.

ONE POSSIBLE WAY TO STRUCTURE A FORECAST REPORT

- Introduction
  Present the problem (what, why, how)
- Literature review
  Optional
- Data
  Source, frequency, main characteristics, stationarity, seasonality, need for transformation, structural breaks etc.
- Methodology – *introduction of methods used, and prior assumptions (what, why, how)*
- Presentation and discussion of results – *tables and graphs with a focus on the interpretation of results and implications*
- Conclusions

No need to include code in the main body, unless you think it's relevant. You are also welcome to include code chunks in the appendix.

THINGS TO CONSIDER

1. The number of observation should be enough to be able to forecast.
2. Data should be original, i.e. do not investigate the unemployment rate in the US or things that we have repeatedly seen in class/workshops/assignments.
3. A "bad" forecast (fx, unreliable large confidence intervals) will not affect your grade as long as your methodology is sound. (However, a bad forecast might indicate that you did something wrong or you did not use a suitable model.)
4. Do not spend too much space explaining the methodology (e.g. what is an AR1 model). Instead, explain **why** the chosen methodology is right for the issue at hand (e.g. the AR1 model can describe this data because it has this and that characteristic). Also, you can/should discuss why your model is preferable to other models (e.g. the ETS model can describe this data better than X because …).
5. All forecast models are equivalent. You do not need to use a VAR with a GETS procedure to get 12. Most forecasts will probably be based on:
   A. ARIMA
   B. ETS
   C. Time series regression
   D. Including dynamic regression
   E. VAR
   - with/without seasonality

METHODOLOGICAL ROUND-UP

THE OVERALL FORECASTING ROUTINE

1. Define (*level, transformations, log-level, differences, growth rates*)
2. Identify (*unit roots, structural breaks, seasonality, deterministic trend*)
3. Estimate
4. Diagnostic checks
5. Forecast
6. Evaluate

Objective:

– to approximate the DGP in a way that leads to the most accurate forecast

AD 1 - TRANSFORMATIONS

Functional

- Logarithms
- Box-Cox
- No clear right/wrong choice

*Ideal: transform to **normal distribution** with stable variance*

Deterministic trend

- Less probable
- But parameters are interpretable
- Can save one differencing(important in small samples)

First differences

- Are (similar to) growth rates
- Widely used

AD 2 - IDENTIFICATION

Is data stationary?

- Unit root → KPSS + ADF tests
- Deterministic trend → ADF test (but low test power)
- Seasonality → Study of correlogram + decomposition
- Structural breaks → QLR *(Quandt Likehood Ratio)* + IS *(Indicator Saturation)*

If not stationary:

- Unit root → differencing
- Deterministic trend → detrending
- Seasonality → seasonal differencing
- Structural breaks → consider modifying the sample period

AD 3 – ESTIMATION

(Classes of) Methods

1. Exponential smoothing (ETS)
2. ARIMA
3. Time series regressions (with GETS as an extension)
4. Dynamic models (with GETS as an extension)
5. VAR

Note:

All method classes need stationary data to valid estimates and identification (except for ETS)

AD 3 – IDENTIFICATION OF ARMA

- Identication is about setting the parameters (p,q), subject to…
- …the principle of **parsimony**

Start by estimating autocorrelation and partial autocorrelation functions:

- **AR(p)**
  - Geometrically (or sinuoisoidal) decaying autocorrelation function after the p'th term.
  - Truncated partial autocorrelation function after the p'th term.
- **MA(q)**
  - Truncated autocorrelation function after the q'th term.
  - Geometrically (or sinuoisoidal) decaying partial autocorrelation function after the q'th term.

Note:
ARIMA() will choose the best model based on information criteria

AD 3 – IDENTIFICATION OF STRUCTURAL BREAKS

- Don’t forget to test for structural breaks!
- In general, recursive Chow test (QLR test)
- Alternatively, IIS/SIS

AD 3 – EXOGENOUS VARIABLES ON THE RIGHT-HAND SIDE

- ARMA models explain variation in current y_t as a function of past values of y_t and some random component.
- However, other variables, x_t, may have an impact on y_t – they should be included to improve precision of estimates.
- Even x_{t-j} may have an effects on y_t, $j = 1,\dots,k$
- Consider including dummies, e.g. to mark outliers or breaks.

We may end up with more variables than observations:
To avoid overfitting ,we should retain only relevant variables.

AD 4 – DIAGNOSTIC CHECKS

- Ljung-Box test is the general way to detect error dynamics
- If residuals have a dynamic, this may affect validity of inference
- Control for error dynamics with a dynamic model, and remember:
  - Data must be stationary
  - We are assuming that **whatever** determines the error dynamics **is <u>independent</u>** from the dynamics of covariates (exogenous variables)

AD 5 – TRAINING AND TEST SUBSETS

- Separate available data into two sets: training and <u>test set</u>
- Use the training set to estimate and choose the best model within a specific class; <u>use the test</u> set to evaluate forecast
- Since the test set is not used in estimation, it provides a reliable indication of how well the model is likely to forecast new data
- Note that “good fit” does not necessarily imply good forecast accuracy!

AD 5 – FORECASTING ACCURACY

- We measure accuracy by summarizing forecast errors:
  - $e_{T+h} = y_{T+h} - \hat{y}_{T+h}$

- Use (one/some of) these measures:
  - MAE (*Mean Absolute Error, easy to interpret*)
  - RMSE (*Root Mean Squared Error, penalizes larger errors more than MAE*)
  - MAPE (*Mean Absolute Percentage Error*)
  - MASE (*Mean Absolute Scaled Error, Scaled MAE, good for comparisons*)
  - CV (*Coefficient of Variation, relative variability of errors*)

Then, choose the forecast to minimize one of these measures.

AD 5 – CROSS VALIDATION

- Instead of one test set, many test sets, ***each consisting of one observation***.
- The training sets are made up of all the previous observations
- Compute the forecast accuracy by averaging across all test sets (evaluation on a rolling forecast origin)
- Choose the model with the lowest (best) accuracy measure obtained in this way

FINAL EXAMPLE

FORECASTING US HOUSING PRICES

1. Load packages
2. Set working directory
3. Load and prepare data
4. Check for possible Box-Cox transformation
5. Preliminary visual analysis
6. Decompose the series
7. Unit-root tests on levels
8. First differencing and stationarity tests
9. Structural break test on first-differenced series
10. Structural break test on post-2008 subsample
11. Re-check stationarity in restricted sample
12. ACF and PACF for model identification
13. Estimate ARIMA and ETS models on original data
14. Inspect model estimates and residual diagnostics
15. Estimate ARIMA and ETS models on Box-Cox transformed data
16. Inspect transformed-data models
17. Ljung-Box residual autocorrelation tests
18. Forecast holdout period
19. Plot forecasts
20. Compare forecast accuracy

See commented R script in [F12_FullProcess.R]

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